Gayatri Vidya Parishad College of Engineering Autonomous

Data Science | Course Code: DS301

Applied Statistics & Data Science - Semester Exam Question Bank & Worked

---

Syllabus Focus: DS301 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Probability Distributions & Sampling Theory

" Syllabus Scope: Discrete and continuous random variables, Normal, Binomial, Poisson, Exponential distri

Unit 2: Parametric Hypothesis Testing

" Syllabus Scope: Null & alternative hypotheses, Type I and Type II errors, p-value interpretation, Z-tes

DS301: Applied Statistics & Data Science - Advanced Topics & Assessment

Academic Track: Data Science - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Categorical Analysis & Chi-Square Tests

" Goodness-of-fit tests, Test of independence in contingency tables, Contingency coefficients, Degrees of free

Unit 4: Analysis of Variance ANOVA

" One-way and Two-way ANOVA, F-distribution, Total sum of squares decomposition, Post-hoc tests Tukey HSD

Unit 5: Correlation & Multivariate Regression

" Pearson and Spearman rank correlation, Multiple linear regression, Multicollinearity, Variance Inflation Fac

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Formulate a two-tailed Student t-test for comparing independent sample means with unequal variances We
2. Construct an ANOVA table for a single-factor experiment with 4 treatment groups and calculate the F-st
3. State and prove the Central Limit Theorem and discuss its application in large-sample statistical infe

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Douglas C. Montgomery - Applied Statistics and Probability for Engineers Wiley 7th Ed
* Gareth James et al. - An Introduction to Statistical Learning Springer

Authorized NoteVault Educational Document " Verified for Registered University Students